Tertiary Syphilis

- **Gummatous syphilis**: Characterized by granulomatous lesions (gummas) presenting as nodules, plaques, or ulcers affecting the skin, mucous membranes, and visceral organs.
- **Late neurosyphilis**: Includes a range of neurological manifestations that may occur years after initial infection:
- Meningeal involvement (chronic meningitis)
- Cranial nerve dysfunction
- Meningovascular syphilis (can lead to stroke or myelitis)
- Parenchymatous neurosyphilis:
  - General paresis (dementia, psychiatric symptoms)
  - Tabes dorsalis (sensory ataxia, lightning pains, areflexia, bladder dysfunction)
- **Cardiovascular syphilis**: Typically manifests decades after infection:
- Aortic regurgitation
- Stenosis of coronary ostia
- Thoracic aortic aneurysm (especially of the ascending aorta)

> **Note**: Neurological involvement in syphilis can occur at any stage—early (e.g., during secondary syphilis) or late (tertiary syphilis).

Laboratory Diagnosis

Demonstration of *Treponema pallidum*

Direct detection methods provide definitive diagnosis of syphilis.

- **Darkfield examination (DFE)**: - Historically the gold standard for diagnosing primary syphilis via visualization of motile spirochetes in fluid from chancres or erosive skin lesions. - Offers immediate results but is labor-intensive, operator-dependent, and prone to false-negative (common) and false-positive (less common) results. - Not recommended for routine use due to the availability of more sensitive and specific molecular methods.

- **Polymerase chain reaction (PCR)**: - Preferred method for direct detection, especially for oral, genital, and cutaneous lesions where contamination with non-pathogenic treponemes may occur. - Can be performed on lesion exudate, tissue biopsies, cerebrospinal fluid (CSF), and blood (though blood PCR is insensitive). - No internationally standardized assay exists; therefore, laboratories must use rigorously validated, quality-controlled PCR protocols.

- **Immunohistochemistry (IHC)**: - Uses polyclonal antibodies against *T. pallidum* to detect treponemes in fixed tissue sections (skin, mucosa, organ biopsies). - Useful in histopathological confirmation but not suitable for routine clinical diagnosis.

- **Hybridization techniques**: - Not used in routine diagnostics due to complexity and limited availability.

- **Warthin-Starry (argentic) staining**: - Difficult to perform and interpret; low sensitivity and specificity. - Rarely useful in clinical practice.

- **Direct fluorescent antibody test (DFA)**: - Requires specialized equipment, reagents, and training. - Not recommended for routine diagnosis.

- **Molecular epidemiological typing**: - Techniques include PCR, PCR-RFLP, multilocus sequence typing (MLST), and whole genome sequencing.

- Limited by the high genetic conservation of *T. pallidum* genome, resulting in low discriminatory power among strains.

Serological Tests for Syphilis (STS)

Serological testing remains essential for the presumptive diagnosis of syphilis.

- **Important limitation**: STS cannot differentiate between venereal syphilis (*T. pallidum* subsp. *pallidum*) and non-venereal treponematoses:
- Yaws (*T. pallidum* subsp. *pertenue*)
- Bejel (endemic syphilis; *T. pallidum* subsp. *endemicum*)
- Pinta (*T. carateum*)
- These organisms are morphologically and antigenically similar. Differentiation relies on clinical presentation, epidemiology, mode of transmission, and genomic sequencing.

> **Clinical implication**: A reactive STS should be managed as syphilis unless there is documented evidence of prior adequate treatment (Grade 1, Level D recommendation).

Non-treponemal tests (NTT)

- Detect antibodies (IgG and IgM) against cardiolipin-lecithin-cholesterol antigen complexes (so-called "reagin" tests).
- Examples:
- Venereal Disease Research Laboratory (VDRL) test
- Rapid Plasma Reagin (RPR) test
- Toluidine Red Unheated Serum Test (TRUST)
- Advantages:
- Inexpensive, simple to perform, and widely available.
- Useful for monitoring disease activity and treatment response (titers correlate with disease activity).
- Seroconversion:
- Typically occurs 10–15 days after appearance of the primary chancre (~6 weeks post-infection).
- Natural history:
- Titers peak 1–2 years after infection.

- Remain detectable at low levels even in late, untreated syphilis.
- Spontaneous seroreversion (conversion to negative) is extremely rare in tertiary syphilis.